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**Problem 1. Mayo (MSK, 2025)**

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The Mayo numerals\*\* are given from 2 to 17 inclusive in a random order. According to the linguists' assumption, the word **busani** historically meant 'greater than N', where N is some other numeral, and another of the numerals presented in the problem meant '2 greater than N':

|                               |                           |
|-------------------------------|---------------------------|
| <b>guoinaiqui</b>             | <b>busani</b>             |
| <b>guoimamnama mamni</b>      | <b>guoimamni</b>          |
| <b>guoimamnama naiqui</b>     | <b>mamni</b>              |
| <b>baji</b>                   | <b>guoimamnama baji</b>   |
| <b>batani</b>                 | <b>guooyi</b>             |
| <b>naiqui</b>                 | <b>guoibusani</b>         |
| <b>guoimamnama guoibusani</b> | <b>guoimamnama guooyi</b> |
| <b>guoimamnama huepu'ulai</b> | <b>guoimamnama busani</b> |

- (a) Determine the correct correspondences.
- (b) What is the value of N?

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**Problem 2. Chibcha (IOLC, 2019)**

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Here's a relation between numbers in Chibcha:

$$\text{gueata} - \text{bosa} = \text{quihicha suhuza} < 19 < \text{quihicha ubchihica} = \text{hisca} \times \text{muihica}$$

- (a) Translate the relation
- (b) Here are some equalities in Chibcha. Write in Hindu-Arabic numerals:

- (1)  $\text{guemuhihas asaqui ata} \div \text{gueatas asaqui cuhupqua} = \text{mica}$
- (2)  $\text{guehiscas asaqui quihicha muihica} \div \text{quihicha aca} = \text{ta}$
- (3)  $\text{gueatas asaqui ata} + \text{guemicas asaqui quihicha ta} = \text{guemuhihas asaqui quihicha cuhupqua}$
- (4)  $\text{gueatas asaqui quihicha hisca} + \text{quihicha mica} = \text{guebosas asaqui suhuza}$

Write in Chibcha:

4, 18, 45, 77, 110, 132, 160, 196

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**Problem 3. Memba (IOLC, 2022)**

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Here are some equations in Memba:

- (1)  $\text{kro}^{35} \times \text{nis}^{55} = \text{c}^{\text{h}}\text{e}^{753} \text{nai}^{35} \text{nai}^{35}$
- (2)  $\text{cen}^{35} \times \text{c}^{\text{h}}\text{e}^{753} \text{nai}^{35} \text{t}^{\text{e}}\text{i}^{53} = \text{c}^{\text{h}}\text{e}^{753} \text{k}^{\text{h}}\text{a}^{55} \text{li}^{53}$
- (3)  $\text{tu}^{31} \text{ku}^{53} \times \text{cen}^{35} = \text{c}^{\text{h}}\text{e}^{753} \text{sum}^{53} \text{t}^{\text{e}}\text{i}^{53} \text{nai}^{35}$
- (4)  $\text{t}^{\text{e}}\text{i}^{53} \text{nis}^{55} + \text{t}^{\text{e}}\text{i}^{53} \text{tu}^{31} \text{ku}^{53} = \text{k}^{\text{h}}\text{a}^{55} \text{li}^{53} \text{t}^{\text{e}}\text{i}^{53} \text{kro}^{35}$
- (5)  $\text{t}^{\text{e}}\text{i}^{53} \text{ko}^{31} \text{ne}^{35} \text{ki}^{31} \text{sum}^{53} \times \text{c}^{\text{h}}\text{e}^{753} \text{pli}^{53} = \text{k}^{\text{h}}\text{a}^{55} \text{li}^{53} \text{pli}^{53}$
- (6)  $\text{t}^{\text{e}}\text{i}^{53} \text{le}^{31} \text{ne}^{53} \times \text{k}^{\text{h}}\text{a}^{55} \text{li}^{53} \text{t}^{\text{h}}\text{e}^{753} = \text{c}^{\text{h}}\text{e}^{753} \text{t}^{\text{e}}\text{i}^{53} \text{le}^{31} \text{ne}^{53} \text{t}^{\text{e}}\text{i}^{53} \text{le}^{31} \text{ne}^{53}$
- (7)  $\text{kro}^{35} \times \text{kro}^{35} \times \text{kro}^{35} + \text{k}^{\text{h}}\text{a}^{55} \text{li}^{53} \text{cen}^{35} \times \text{k}^{\text{h}}\text{a}^{55} \text{li}^{53} \text{cen}^{35} = \text{t}^{\text{e}}\text{o}^{55}$
- (8)  $\text{k}^{\text{h}}\text{a}^{55} \text{li}^{53} \text{le}^{31} \text{ne}^{53} \div \text{c}^{\text{h}}\text{e}^{753} \text{le}^{31} \text{ne}^{53} = \text{x}$
- (9)  $\text{c}^{\text{h}}\text{e}^{753} \text{sum}^{53} \text{nai}^{35} - \text{k}^{\text{h}}\text{a}^{55} \text{li}^{53} \text{t}^{\text{e}}\text{i}^{53} \text{nai}^{35} = \text{y}$

(a) Write the equations in digits.

(b) Write in Memba:

16, 23, 48, 77, 111, 210

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**Problem 4. Tongan (JOL, ???)**

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Here are some equations in ???:

- (1)  $\text{valu ono} = 86$
- (2)  $\text{tolu} \times \text{tolu tolu} = \text{hiva hiva}$
- (3)  $\text{taha tolu} \times \text{fitu noa} = \text{hiva taha noa}$
- (4)  $\text{ua nima} \times \text{fā} = \text{taha noa noa}$

(a) Write in Tongan: 324, 1228

(b) Write in digits: **fita ua valu**

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**Problem 5. Old Uyghur (MSK, 2022)**

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Here are some numbers in Old Uyghur and their translations in random order:

**altı, beş tokuz on, bir otuz, bir yegirmi, kırk sekiz, on altı, otuz beş, sekiz on beş, tokuz tokuz on, yegirmi altı, yegirmi bir**

6, 11, 16, 21, 26, 35, 48, 85, 89

(a) Some numbers have multiple forms, but not all that do are translated. Write the remaining forms.

(b) Match the numbers.

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**Problem 6. Lele (MSK, 2017)**

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Arithmetic examples are given in the Lele language\*. All numbers in the examples are less than 100:

- (1) **manonoh** × **manondoruoh** = **mahangul e ondoruoh**
- (2) **manondrsih** × **manondrsih** = **manondorungul e sih**
- (3) **manonoh** × **masungul** = **manonngul**
- (4) **matoloh** + **marngul e ruoh** = **marngul e limah**
- (5) **malimngul e onoh** + **masungul e hahou** = **manondrtilngul**

(a) Write in Lele:

2, 7, 65, 99

(b) Write in numbers:

(6) **matilngul e tolroh** + **mahahou** = **matilngul e ondrtoloh**

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**Problem 7. Tlapanec (Onling, 2023)**

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Here are some equalities in Tlapanec:

- (1) **guwa-isu-mu** × **ajku** = **ajku-rumi ijma-mo**
- (2) **mijna-guwa** × **ajma** = **guwa-nisu isu-mu**
- (3) **mba-skiyun emba-mo** + **guwa-nisu ijma-mo** = **mba-skiyun guwa mingiyun**
- (4) **ajsu!** = **majju**
- (5) **mba-skiyun mijna-guwa** × **ajsu** = **mba-jku-skiyun juan**
- (6) **guwa-ijma-mo** × **ajsu** = **mba-skiyun guwa majju**
- (7) **ajku** × **ajku** = **guwa-nisu emba-mo**
- (8) **mba-skiyun guwa-isu-mu** + **[A]** = **mba-jma-skiyun ijku-mu**
- (9) **mba-skiyun wisu** + **[B]** = **mba-skiyun guwa wisu**
- (10) **[C]** × **ajma** = **mba-jsu-skiyun**

(a) Write the equalities 1-10 in Arabic numerals

(b) Write the Tlapanec numerals corresponding to A, B, and C.

(c) Write in Tlapanec:

19, 53, 69, 77, 88

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**Problem 8. Skou (MSK, 2015)**

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Here are some equations in Skow:

- (1)  $\text{hangpà pa hìngtung} + \text{náhìpa} = 22$
  - (2)  $\text{hangpà pa náhìpa pa héngtong} - \text{náhìpa pa hìngtung} = 13$
  - (3)  $\text{hangpà pa ná pang pa áling} + \text{héngtong} = 21$
  - (4)  $\text{hangpà pa náhìpa} - \text{hangpà pa nongpong} = 4$
  - (5)  $\text{hìngtung} + \text{áling} = \text{héngtong}$
  - (6)  $\text{nongpong} + \text{héngtong} = \text{náhìpa} - \text{áling}$
- (a) Write the equations in digits.
- (b) Write in digits:  $\text{hangpà pa héngtong}$ ,  $\text{náhìpa pa nongpong}$ ,  $\text{hangpà pa ná pang}$
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**Problem 9. Albanian (KrLO, ???)**

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Here are some equations in Albanian:

- (1)  $\text{shtatëmbëdhjetë} + \text{njëzet e nëndë} = \text{dizet e gjashtë}$
  - (2)  $\text{diqint e gjashtëdhjetë} - \text{nëndëdhjetë e tri} = \text{njëqint e trizet e shtatë}$
  - (3)  $\text{tetëdhjetë e një} \times \text{katrëmbëdhjetë} = \text{njëmiljë e njëqint e tridhjetë e katër}$
  - (4)  $\text{katrëzet e pesë} \div \text{pesë} = 17$
- (a) Write the equations in digits
- (b) Write the numbers in Albanian: 8, 16, 350, 2049
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**Problem 10. Ainu (Sweden, 2022)**

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Here are some equations in Ainu:

- (1)  $\text{iwan} + \text{arwan} = \text{re ikasma wan}$
  - (2)  $\text{re} \times \text{ine ikasma wan} = \text{tu ikasma tu hotne}$
  - (3)  $\text{sine} + \text{ine} = \text{tu} + \text{re}$
  - (4)  $\text{re} \times \text{re ikasma wan} = \text{sinepesan ikasma wan etu hotne}$
  - (5)  $\text{asikne} \times \text{asikne ikasma wan} = \text{asikne ikasma wan eine hotne}$
  - (6)  $\text{tu}^{\text{re}} + \text{re}^{\text{tu}} = \text{arwan ikasma wan}$
  - (7)  $\text{ine} \times \text{re ikasma wan} = \text{tu ikasma wan ere hotne}$
  - (8)  $\text{sinepesan} \times \text{re} = \text{arwan ikasma hotne}$
  - (9)  $\text{sine} + \text{tupesan}^{\text{tu}} = \text{asikne ikasma re hotne}$
  - (10)  $\text{ine ikasma ine hotne} + \text{tu ikasma wan} = \text{iwan ikasma wan easikne hotne}$
- (a) Write the equations in digits
- (b) Write in digits:  
 $\text{tu}$ ,  $\text{hotne}$ ,  $\text{wan earwan hotne}$ ,  $\text{sinepesan ikasma wan}$ ,  $\text{ine ikasma wan eiwan hotne}$
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**Problem 11. Nimbishka+Karshishka (Taiwan, 2020)**

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Here are some numbers in the Nimbishka dialect of Guandala. All numbers are less than 250:

- (1)  $\text{bo'o}^{\text{bi}} = \text{gume furu ni da}$
- (2)  $\text{da} + \text{biyar} + \text{furu} = \text{bi}^{\text{bi}} + \text{shide}$
- (3)  $\text{gume bi ni kwada} = \text{biyar} \times \text{bo'o}$
- (4)  $\text{ugu}^{\text{biyar}} = \text{wo ni gume tager ni ugu}$
- (5)  $\text{wo ni ugu} - \text{tager} = \text{gume kwada ni bo'o}$
- (6)  $\text{gume bo'o ni tager} \div \text{furu} = \text{tuni mbe kwada}$
- (7)  $\text{tuni mbe shide} + \text{da}^{\text{shide}} + \text{gwom} = \text{gume bi ni biyar}$
- (8)  $\text{tuni mbe gwom} + \text{gume shide ni ugu} = \text{gume tager ni da}$
- (9)  $\text{wo ni gume biyar ni bi} = \text{gume shide ni furu} + \text{gume gwom ni gwom}$

(a) Write in Karshishka dialect: 20, 100, 110, 153, 999

(b) Here are some equalities between the Nimbishka(LHS) and Karshishka(RHS) dialect, write in digits:

- (10)  $\text{gom sha takushi} + \text{gom sha biyari} = \text{gume bi ni tanran}$
- (11)  $\text{ashirini n huru} + \text{deri bi} = \text{wo ni gume shide ni tager}$
- (12)  $\text{gom sha tara} + \text{bakwe} + \text{ashirini n uku} = \text{tuni mbe tanran} + \text{gume bi ni furu}$

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**Problem 12. Oboro (ČLO, 2017)**

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Here are some equalities in Oboro (traditional and new). All numbers are between 11 and 99 inclusive:

- (1)  $\text{akəp me iba} = \text{akəp me iba}$
- (2)  $\text{akəp me jeeta} = \text{inaan iba}$
- (3)  $\text{akəp iba me go} = \text{inaan go me etip me akəp}$
- (4)  $\text{akəp ita me ini} = \text{etip me akəp me ini}$
- (5)  $\text{akəp ita me jaaba} = \text{inaan ita me etip iba}$
- (6)  $\text{akəp ini me ge} = \text{etip iba me ge}$
- (7)  $\text{akəp ini me jaaba} = \text{inaan ita me etip iba me akəp}$
- (8)  $\text{akəp go me iba} = \text{etip iba me akəp me iba}$
- (9)  $\text{akəp go me onaange} = \text{inaan ge me etip ita}$
- (10)  $\text{akəp gweregwen} = \text{etip ita}$
- (11)  $\text{akəp jaaba me jeeta} = \text{inaan iba me etip ini}$
- (12)  $\text{akəp onaange me go} = \text{inaan go me etip go}$

(a) Write the equalities in digits

(b) Write in digits:  $\text{akəp me go}$ ,  $\text{akəp gweregwen me ita}$ ,  $\text{akəp jeeta me gweregwen}$

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**Problem 13. Tifal (MSK, 2017)**

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Here are some equations in Tifal. All numbers are less than 30:

- (1)  $\text{asumano} \times \text{aleeb} = \text{bokob}$
- (2)  $\text{asumano} \times \text{ataling} = \text{tadang}$
- (3)  $\text{bokob} \times \text{ataling} = \text{ataling madi}$
- (4)  $\text{bokob} \times \text{asumano} = \text{nakal madi}$
- (5)  $\text{asumano} \times \text{feet} = \text{feet madi}$
- (6)  $\text{ataling} \times \text{ataling} = \text{tadang madi}$
- (7)  $\text{asumano} + \text{ataling} = \text{feet}$
- (8)  $\text{feet} + \text{miit} = \text{feet madi}$
- (9)  $\text{tadang} + \text{ataling} = \text{tadang madi}$
- (10)  $\text{beeti} + \text{nakal} = \text{beeti madi}$
- (11)  $\text{bokob} + \text{maakob} = \text{feet}$
- (12)  $\text{awok} \times \text{awok} = \text{asumano madi}$

(a) Write the equations in digits

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**Problem 14. Sora (Bulgaria, 2012)**

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Here are some equations in Sora. All numbers are less than 200:

- (1)  $\text{unji} + \text{gulji} = \text{gəlmuy}$
- (2)  $\text{gəlmuy} + \text{yagi} = \text{mizgəl-bagu}$
- (3)  $\text{tinji} + \text{tamji} = \text{mizgəl-mənləy}$
- (4)  $\text{gəlji} + \text{bakəri gəlji} = ?$
- (5)  $\text{bakəri mizgəl-yagi} + \text{mizgəl-tudru} = \text{yakəri mizgəl-bəy}$
- (6)  $\text{gulji} \cdot \text{bagu} = \text{mizgəl-bagu}$
- (7)  $\text{bakəri mizgəl-mənləy} \cdot \text{yagi} = ?$
- (8)  $\text{unji} \cdot \text{unji} = \text{mizgəl-unji}$
- (9)  $\text{bakəri tamji} \cdot \text{bagu} = \text{unjikəri mizgəl-unji}$
- (10)  $\text{gəlji} \cdot \text{mizgəl} = ?$
- (11)  $\text{mizgəl-bəy} \cdot \text{mizgəl-bəy} = \text{tamjikəri tinji}$

(a) Write the equations in digits

(b) Write the digits corresponding to ?

(c) Write in Sora: 228, 138

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**Problem 15. SL (UkrLO, 2020)**

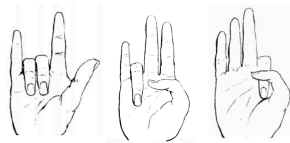
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Here are some numbers and their sign-language representations in random order:

5, 9, 21, 27, 50, 53, 70



(a) Make gang signs for: 2, 20, 57, 78



(b) Write in digits: